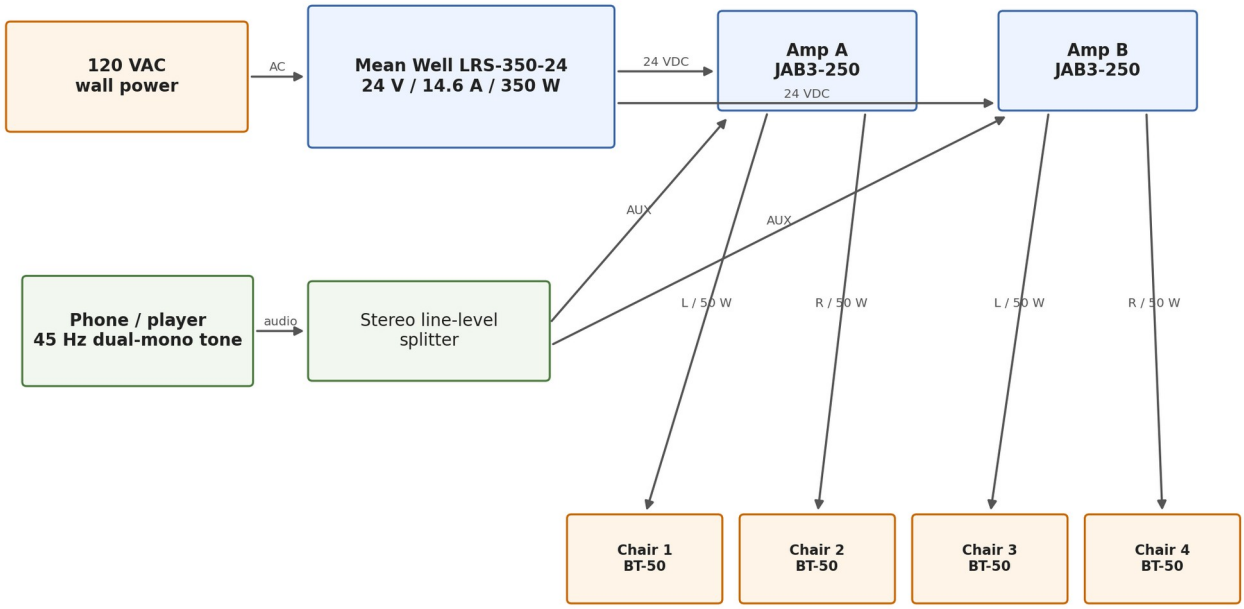
FOUR-CHAIR HALLOWEEN VIBRATION PROP

Build & wiring guide

4 independent folding chairs • 4 × 50 W bass shakers • 2 × stereo Class-D amps



Only passive speaker wire and the shaker need to reach each chair.

Design goal: A sudden, strong low-frequency vibration in each chair while keeping all mains voltage and active electronics in one remote controller box. Only passive speaker wire and a bass transducer go to each chair.

Revision: October 2026

1. Final design at a glance

The four-chair system uses one 50 W / 4 Ω MCM BT-50 bass transducer per chair. Two SURE-JAB3-250 stereo amplifier boards provide four independent 50 W channels. A single Mean Well LRS-350-24 supplies 24 VDC to both amplifier boards. A phone or other audio source sends the same low-frequency signal to both amps.

Why this combination works: The two amplifiers require about 130 W each at rated output (260 W total nominal), while the LRS-350-24 can supply 350.4 W / 14.6 A at 24 V. Each shaker gets its own 4 Ω amplifier channel; no series or parallel speaker wiring is needed.

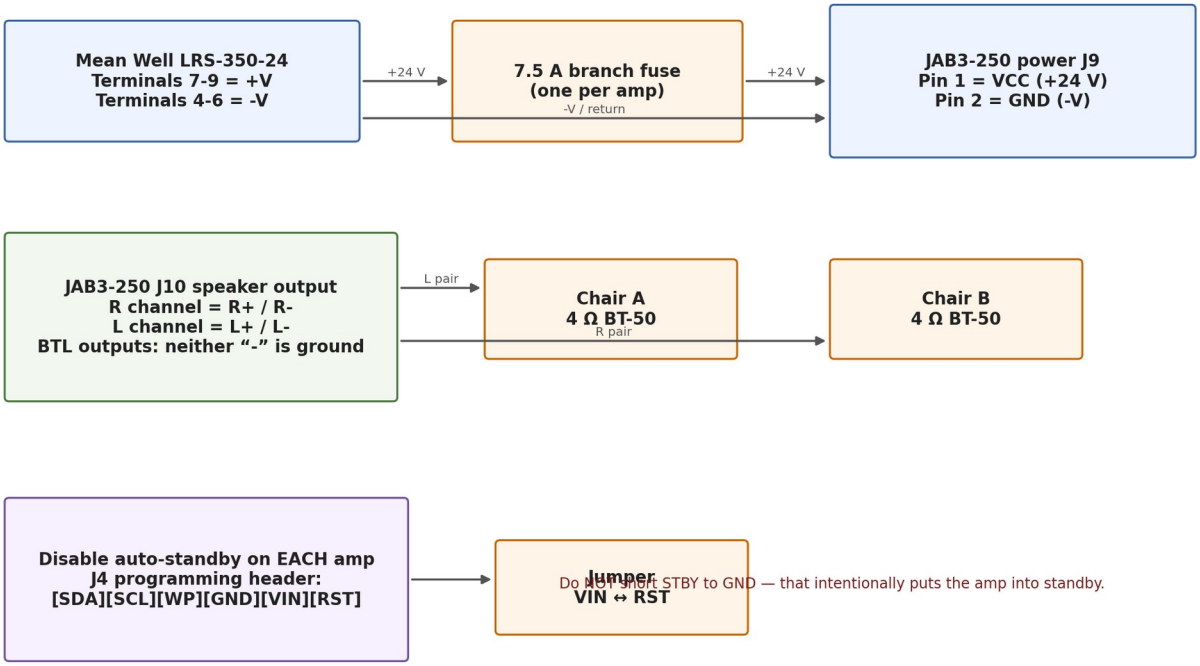
Core parts

Qty	Part	Role / key spec	Source / part no.
4	MCM Custom Audio BT-50	Bass transducer; 4 Ω; 50 W RMS; 45 Hz resonance; 28–55 Hz range	Newark 71Y5716
2	Multicomp Pro SURE-JAB3-250	2 × 50 W @ 4 Ω Class-D amplifier board with ADAU1701 DSP	Newark 61AK5098
1	Mean Well LRS-350-24	24 VDC, 14.6 A, 350.4 W AC/DC switching supply	Amazon / Mean Well
1	Namunanee 16 × 12 × 5.9 in ABS enclosure	Thermostat, fan, internal mounting panel, hinged cover	Amazon; selected model

Small parts / consumables

- 16 AWG stranded speaker wire: one two-conductor run from the controller to each chair. 14 AWG is fine for unusually long runs.
- 16 AWG red/black stranded wire for 24 VDC amp power branches; 14–16 AWG for the short PSU feed if using a distribution point.
- 2 × inline automotive blade-fuse holders + 2 × 7.5 A fuses, one fuse in the +24 V feed to each amplifier board.
- Stereo 3.5 mm line-level splitter plus two AUX leads/adapters that match the included JAB3 AUX harnesses.
- Two short insulated jumper wires (or suitable 2.0 mm-pitch connector leads) to jumper VIN↔RST on J4 of each amplifier.
- Crimp terminals / ferrules / heat-shrink as appropriate; do not leave loose strands at screw terminals.
- Per chair: rigid mounting plate (about 6–8 in square, 3/4 in plywood is easy), shaker mounting hardware, and U-bolts or strong clamps to tie the plate to the metal chair frame.
- Cable protection: gaffer tape, floor cord covers, or routing that eliminates trip hazards.

2. Electrical wiring



For the J10 physical pin order, verify the supplied harness/board before energizing; see the note in the guide.

2.1 AC mains → Mean Well power supply

For U.S. 120 V operation, set the LRS-350 input-voltage selector to 115 V before connecting power. The Mean Well terminal assignment is:

Mean Well terminal	Function	Connection
1	AC/L	Hot / line (normally black in a U.S. grounded cord)
2	AC/N	Neutral (normally white)
3	FG	Protective earth / ground (normally green)
4–6	-V	24 VDC negative / return
7–9	+V	24 VDC positive

Before connecting either amplifier: Power the Mean Well by itself and measure between +V and -V with a multimeter. Adjust V ADJ to 24.0 VDC. The JAB3-250 is rated for 12–26 V; do not "turn it up" above 24 V to get more shake.

2.2 Mean Well → both amplifier boards

Wire the two amplifier boards in parallel from the 24 V supply. The Mean Well's multiple +V terminals are internally common, as are the multiple -V terminals.

- Mean Well +V terminal 7 → 7.5 A fuse → Amp A J9 pin 1 (VCC).
- Mean Well -V terminal 4 → Amp A J9 pin 2 (GND).
- Mean Well +V terminal 8 → 7.5 A fuse → Amp B J9 pin 1 (VCC).
- Mean Well -V terminal 5 → Amp B J9 pin 2 (GND).

Leave the remaining Mean Well +V/-V terminals available for future expansion or the enclosure fan/control circuitry if the enclosure documentation specifically calls for 24 VDC. Do not assume the enclosure fan voltage; follow its supplied wiring instructions.

2.3 Amplifier channels → four chairs

Assign one amplifier channel to each shaker:

Amplifier	Channel	Chair	Load
Amp A	Left	Chair 1	BT-50, 4 Ω
Amp A	Right	Chair 2	BT-50, 4 Ω
Amp B	Left	Chair 3	BT-50, 4 Ω
Amp B	Right	Chair 4	BT-50, 4 Ω

Critical BTL-output rule: each shaker gets its own + and - pair from one amplifier channel. Do not connect either speaker "-" lead to PSU ground, chassis/earth, or another channel's "-" lead. Do not tie the four speaker negatives together.

J10 physical pin order: the SURE/Multicomp documentation names J10 as the four-pin speaker connector but does not print the pin order in its current five-page datasheet. Other JAB3-family connection documentation uses the following order:

```
J10 pin 1 = Right +
J10 pin 2 = Left  -
J10 pin 3 = Right -
J10 pin 4 = Left  +
```

Verification step: Before energizing the finished build, verify the supplied J10 harness/board markings against the board you receive. Wire by channel pair (R+/R- and L+/L-), not by assumed wire color. This is the one connector detail worth confirming physically because the exact Multicomp datasheet omits the pin-order table.

3. Audio signal, standby fix, and amp settings

3.1 Feed the same signal to both amplifier boards

Use one source and split its stereo line-level output into two identical feeds: one to Amp A AUX IN and one to Amp B AUX IN. The trigger audio should be dual-mono: the same low-frequency waveform in the left and right channels, so all four chairs fire together.

```
Phone / player
   ↓
USB-C/Lightning audio adapter (if needed)
   ↓
Stereo line-level Y splitter
   ├──→ Amp A AUX IN
   └──→ Amp B AUX IN
```

3.2 Disable the automatic signal-sensing standby

The JAB3 normally enters low-power standby after about one minute without detected audio. Sure Electronics provides an official configuration for disabling this behavior: short VIN and RST on the J4 programming connector. Do this on both amplifier boards.

```
J4: [SDA] [SCL] [WP] [GND] [VIN] [RST]
                                 └──── jumper ────┘
```

Do not use J12 for this: J12 pin 1 is STBY and pin 2 is GND. Shorting STBY→GND intentionally puts the amp into standby. J12 pin 3 (MUTE) should not be shorted to GND when the JAB3 is used by itself.

3.3 Initial amplifier settings

The exact SURE-JAB3-250 / Multicomp datasheet assigns the speaker controls as follows in stereo mode:

Control	Function	Starting point for this project
POT1	Speaker-output gain	Start around midrange; raise only if needed.
POT2	Speaker high-pass filter (20 Hz–2 kHz)	Set to the lowest cutoff (~20 Hz) so 40–45 Hz reaches the shakers.
POT3	Headphone-output high-pass filter	Not important for this build.
POT4	Overall volume	Start low; use this for controlled commissioning.

The datasheet does not specify knob direction for minimum/maximum cutoff. Confirm POT2 by playing a 45 Hz tone at low level and adjusting until the shaker output is strongest, then leave both boards matched.

3.4 Trigger waveform

The BT-50's published resonance is 45 Hz, so 40–45 Hz is the best starting range. For a startling effect, abrupt short bursts are more effective than leaving a continuous vibration running.

- Commissioning tone: continuous 45 Hz sine at low volume.
- Scare pattern starting point: 0.6–0.8 s at ~45 Hz, 0.15–0.25 s silence, then a 0.3–0.5 s second burst.
- Use a clean signal; if the amplifier is audibly clipping or the shaker is making harsh mechanical noises, back the level down.

4. Chair installation

Use one BT-50 per folding chair. The goal is to couple the shaker into the chair frame, not merely buzz a flexible patch of plastic.

- Mount the BT-50 rigidly to a small 3/4 in plywood plate (roughly 6–8 in square; size to the actual mounting holes).
- Clamp or U-bolt that plate firmly across two structural metal frame members under the seat. Avoid the folding hinge, locking mechanism, and any tubing that flexes excessively.
- Keep the shaker and its moving parts completely inaccessible to fingers, clothing, and dangling costume pieces.
- Secure the speaker wire to the chair frame with strain relief so a tug on the floor cable is not transferred to the shaker terminals.
- Use intact non-slip chair feet. If the chair begins walking/skittering across the floor, reduce level rather than trying to hold it in place with more aggressive restraints.

Mechanical rule: The chair must remain a chair. Do not drill or clamp through structural parts in a way that weakens the frame or defeats its locking mechanism. The desired effect is intense vibration, not enough motion to unseat someone.

5. Bring-up and commissioning sequence

1. Complete all mechanical chair mounting and cable strain relief with the controller unplugged.
2. With the amps disconnected, set the Mean Well selector to 115 V (U.S.), power the supply, and adjust the output to 24.0 VDC with a multimeter. Unplug it again.
3. Connect Amp A and Amp B power through their individual 7.5 A +24 V fuses. Check polarity twice before power-up.
4. Install the VIN↔RST J4 jumper on both amplifier boards to disable auto-standby.
5. Connect only Chair 1 to Amp A Left. Leave source and amplifier volume low.
6. Play a 45 Hz dual-mono tone. Slowly raise POT4 / source level until Chair 1 produces a strong but controlled vibration. Adjust POT2 to the lowest speaker HPF cutoff and match POT1 gain as needed.
7. Connect Chair 2, then Chairs 3 and 4, verifying each channel individually before testing all four simultaneously.
8. Match Amp A and Amp B volume/gain by sitting in or mechanically observing each chair at the same source level.
9. Stop the signal for several minutes, then trigger it again. All four chairs should respond immediately; this confirms the signal-sensor standby bypass is working.
10. Run a rehearsal with repeated short bursts for 10–15 minutes. Check PSU, fuse holders, connectors, amp boards, and wiring for abnormal heat or odor. Warm is normal; hot connectors or discoloration are not.

Recommended operating setup

- Controller box sits away from the chairs; keep the Mean Well and both amplifier boards together.
- Long runs are speaker wire, not long high-current 24 V runs.
- Route/tape/cover all floor runs so nobody can catch a foot on them.
- If outdoors or anywhere damp, use GFCI-protected power and keep the electronics/enclosure within the conditions its manufacturer allows.

6. Do-not-do list

- Do not wire two BT-50 shakers in parallel on one channel. Each chair already has its own 50 W / 4 Ω channel.
- Do not connect speaker-output negatives together or to ground. The Class-D outputs are bridge-tied.
- Do not run 120 VAC to any chair. Mains stays at the controller.
- Do not operate the Mean Well with accessible mains terminals. Bench setup is for commissioning only; the final system must have mains connections enclosed and strain-relieved.
- Do not raise the Mean Well above 24 V to chase more vibration. The amplifier's maximum rated operating voltage is 26 V.
- Do not rely on a color of the J10 speaker harness until you have verified which conductor belongs to each +/- output.
- Do not use the effect on anyone whose fall/medical risk makes a sudden startle inappropriate. Test the final intensity on yourself first and keep the chair stable.

7. Labeling scheme

A little labeling will make setup and teardown much easier. Use the same label at both ends of every run:

Label	Amp/channel	Destination	Suggested cable ID
C1	Amp A / Left	Chair 1	C1
C2	Amp A / Right	Chair 2	C2

C3	Amp B / Left	Chair 3	C3
C4	Amp B / Right	Chair 4	C4
AUX-A	Audio split A	Amp A	AUX-A
AUX-B	Audio split B	Amp B	AUX-B

8. Verified technical references

The electrical details in this guide were checked against the following manufacturer/distributor documentation:

- Mean Well LRS-350 series datasheet — 24 V / 14.6 A / 350.4 W model data; 115/230 V selector; terminal assignment; output adjustment range; forced-air fan.
- Multicomp Pro SURE-JAB3-250 at Newark — Exact amplifier purchased: 2 × 50 W, 4 Ω, JAB3 Series, ADAU1701 DSP.
- Multicomp Pro SURE-JAB3-250 datasheet — 24 V nominal / 12–26 V range; 130 W nominal requirement; J9 power pinout; J4 programming header; standby behavior; potentiometer functions.
- Sure Electronics JAB3 configuration note — Official instruction to short J4 VIN and RST to turn off the signal-level sensor system.
- MCM Custom Audio BT-50 at Newark — 4 Ω, 50 W RMS, 45 Hz resonance, 28–55 Hz operating range, 3 in weighted plate.

One documentation caveat: The exact Multicomp SURE-JAB3-250 datasheet identifies J10 as the speaker-output connector but omits its physical pin-order table. This guide therefore calls for a physical check of the supplied J10 harness/board before power-up. The channel-pair rule itself is not ambiguous: each shaker must connect only across one amplifier channel's + and - outputs.

Quick wiring card

```
AC / PSU
  120 VAC hot   → LRS terminal 1 (AC/L)
  120 VAC neutral→ LRS terminal 2 (AC/N)
  Earth/ground   → LRS terminal 3 (FG)
  Set selector: 115 V; adjust output: 24.0 VDC

24 V POWER
  LRS +V → 7.5 A fuse → Amp A J9 pin 1 (VCC)
  LRS -V              → Amp A J9 pin 2 (GND)
  LRS +V → 7.5 A fuse → Amp B J9 pin 1 (VCC)
  LRS -V              → Amp B J9 pin 2 (GND)

CHAIRS
  Amp A Left  → Chair 1 BT-50
  Amp A Right → Chair 2 BT-50
  Amp B Left  → Chair 3 BT-50
  Amp B Right → Chair 4 BT-50
  Never tie speaker negatives together or to ground.

AUTO-STANDBY BYPASS
  On EACH amp: J4 VIN ↔ RST jumper

AUDIO
  Dual-mono 40–45 Hz source → stereo splitter → AUX A + AUX B
```

```
AMP STARTING SETTINGS
  POT2 speaker HPF → lowest cutoff (~20 Hz)
  POT4 volume → low, then raise gradually
  POT1 gain → start around midrange
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